Experiment 9

Student Name: Abhinav Kumar Prasad
UID: 24BAI70356
Branch: CSE - AIML
Section/Group: 24AIT_KRG G1
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Aim

To create and implement PL/SQL packages by developing a package specification and package body containing procedures and shared cursors, in order to achieve modular, reusable, and efficient database programming.

Software Requirements

- Database Management System:
  - Oracle Database Express Edition (Oracle XE)
  - PostgreSQL Database
- Database Administration Tool / Client Tool:
  - Oracle SQL Developer (for Oracle XE)
  - pgAdmin (for PostgreSQL)

Objectives

- To design and implement a PL/SQL package that includes procedures and shared cursors for structured and modular program development.

Problem Statement

- In enterprise applications, related database operations should be grouped together for better management and reuse.

Practical/Experiment Steps

- Modular Architecture Design: Structured the database logic into a two-part PL/SQL package consisting of a public specification and a private implementation body.

- Encapsulation of Global Assets: Defined a shared cursor within the package body to provide a centralized and reusable data source for multiple procedures.
- Procedural Logic Development: Authored specialized procedures to handle varied data retrieval tasks, including full table iteration and targeted record searching.
- Information Hiding Implementation: Enforced security and abstraction by hiding the internal cursor logic from the end-user while exposing only necessary procedure calls.
- Integration Testing: Executed the package subprograms within anonymous blocks to validate cross-procedure consistency and data accuracy.

Procedure

- Established a connection to the Oracle/PostgreSQL environment and enabled the output console for result verification.
- Created the employees table and populated it with a baseline dataset to facilitate testing.
- Drafted the Package Specification (emp_package AS) to declare the public interface of the procedures.
- Constructed the Package Body (emp_package AS BODY) to house the shared emp_cursor and the actual logic for the declared procedures.
- Implemented a FOR loop within show_employees to automatically handle the opening, fetching, and closing of the shared cursor.
- Developed the logic for get_employee to filter the shared cursor data based on a user-provided input parameter.
- Compiled both the specification and body to ensure there were no syntax errors or dependency issues.
- Invoked emp_package.show_employees to test global data retrieval and printed the formatted output.
- Called emp_package.get_employee(2) to verify the ability of the package to handle specific conditional requests using shared resources.

Input/Output Analysis

SQL Input Queries

```
CREATE TABLE employees (
   emp_id NUMBER PRIMARY KEY,
   emp_name VARCHAR2(50),
```

```
    salary NUMBER
);
```

Output

Query result | **Script output** | DBMS output | Explain Plan | SQL history

```
SQL> CREATE TABLE employees (
         emp_id NUMBER PRIMARY KEY,
         emp_name VARCHAR2(50),
         salary NUMBER...
Show more...


Table EMPLOYEES created.

Elapsed: 00:00:00.020
```

SQL Input Queries

```
INSERT INTO employees VALUES (1, 'Amit', 30000);
INSERT INTO employees VALUES (2, 'Riya', 40000);
INSERT INTO employees VALUES (3, 'John', 50000);

COMMIT;
```

Output

Query result | **Script output** | DBMS output | Explain Plan | SQL history

```
SQL> INSERT INTO employees VALUES (1, 'Amit', 30000)


1 row inserted.

Elapsed: 00:00:00.019

SQL> INSERT INTO employees VALUES (2, 'Riya', 40000)


1 row inserted.

Elapsed: 00:00:00.002

SQL> INSERT INTO employees VALUES (3, 'John', 50000)


1 row inserted.

Elapsed: 00:00:00.001
```

```
SQL> COMMIT


Commit complete.

Elapsed: 00:00:00.002
```

SQL Input Queries

```
CREATE OR REPLACE PACKAGE emp_package AS
   -- Procedure to display all employee details
   PROCEDURE show_employees;

   -- Procedure to display employee by ID
   PROCEDURE get_employee(p_id NUMBER);
END emp_package;
/
```

Output

Query result | **Script output** | DBMS output | Explain Plan | SQL history

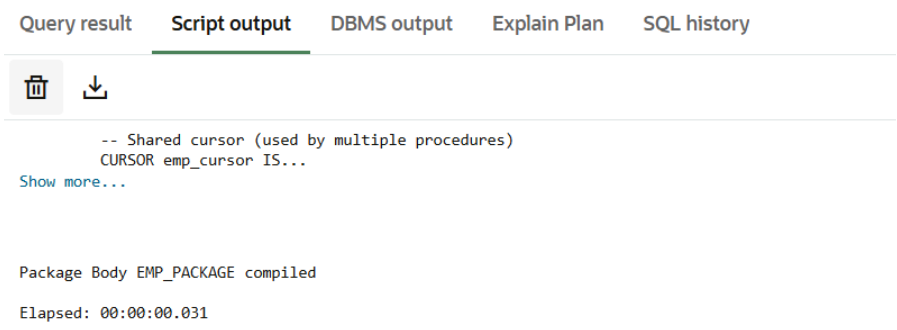
```
        -- Procedure to display all employee details
        PROCEDURE show_employees;
    ...
Show more...


Package EMP_PACKAGE compiled

Elapsed: 00:00:00.011
```

SQL Input Queries

```
CREATE OR REPLACE PACKAGE BODY emp_package AS

   CURSOR emp_cursor IS
      SELECT emp_id, emp_name, salary FROM employees;


   PROCEDURE show_employees IS
   BEGIN
      FOR rec IN emp_cursor LOOP
         DBMS_OUTPUT.PUT_LINE('ID: ' || rec.emp_id ||
                              ' Name: ' || rec.emp_name ||
                              ' Salary: ' || rec.salary);
```

```
    END LOOP;
  END;


  PROCEDURE get_employee(p_id NUMBER) IS
  BEGIN
    FOR rec IN emp_cursor LOOP
      IF rec.emp_id = p_id THEN
        DBMS_OUTPUT.PUT_LINE('Employee Found -> ID: ' || rec.emp_id ||
                             ' Name: ' || rec.emp_name ||
                             ' Salary: ' || rec.salary);
      END IF;
    END LOOP;
  END;

END emp_package;
/
```

Output



SQL Input Queries

```
SET SERVEROUTPUT ON;


BEGIN
  emp_package.show_employees;
END;
/
CALL update_student(1, 'Rahul Verma', 22, 'Data Science');
```

Output

Query result | Script output | DBMS output | Explain Plan | SQL history

```
ID: 1 Name: Amit Salary: 30000
ID: 2 Name: Riya Salary: 40000
ID: 3 Name: John Salary: 50000


PL/SQL procedure successfully completed.

Elapsed: 00:00:00.128
```

SQL Input Queries

```
BEGIN
   emp_package.get_employee(2);
END;
/
```

Output

Query result | Script output | DBMS output | Explain Plan | SQL history

```
    END;


Employee Found -> ID: 2 Name: Riya Salary: 40000


PL/SQL procedure successfully completed.

Elapsed: 00:00:00.123
```

Learning Outcomes

- Package Structure Proficiency: Gained a clear understanding of the separation between the package specification (header) and the package body (implementation).
- Modular Programming Mastery: Learned to group related database operations into a single logical unit to improve maintainability and code organization.
- Shared Resource Management: Expertise in utilizing shared cursors to reduce memory overhead and ensure consistent data fetching across multiple subprograms.

- Encapsulation and Security: Understood how packages provide a layer of abstraction by hiding complex implementation details from the calling environment.